

Table 1: Comparison of different paraphrasing attacks against 8 text-detection algorithms (at 5% FPR) using GPT-3.5-turbo generated texts from three different datasets. The best performances are bolded.

Dataset	Attack	Sim	Defense								Avg.
			LogRank	DetectGPT	DNA-GPT	Fast-DetectGPT	Raidar	TOCSIN	RoBERTa	R-Detect	
XSum	No Attack	-	63.33	26.67	80.00	95.33	28.75	98.00	66.67	69.67	66.05
	Dipper	86.67	15.67	2.67	27.33	76.33	13.75	74.67	86.67	48.00	43.14
	Raidar	100.00	49.00	12.00	22.64	84.67	16.25	90.67	55.33	68.67	49.90
	Ours	94.00	4.67	4.00	21.33	17.00	0.00	26.67	22.67	4.67	12.63
SQuAD	No Attack	-	67.00	12.67	41.33	93.50	22.00	92.67	41.33	79.67	56.27
	Dipper	75.33	23.67	2.67	10.00	77.67	5.00	75.33	67.33	53.33	39.38
	Raidar	95.33	58.00	12.33	19.33	81.67	26.00	84.00	32.67	73.50	48.44
	Ours	88.67	8.33	2.67	8.67	27.50	5.00	25.33	7.33	3.67	11.06
LongQA	No Attack	-	73.83	33.33	10.67	86.00	36.25	88.67	38.67	89.33	57.09
	Dipper	94.67	28.83	6.00	0.67	75.00	5.00	67.33	65.33	74.00	40.27
	Raidar	100.00	59.33	22.67	1.33	72.67	33.75	77.33	28.00	79.50	46.82
	Ours	95.33	14.50	5.00	0.00	11.33	0.00	16.00	6.67	6.00	7.44

LLM-Generated paragraph	Dipper paraphrase	CoPA
The Korea Space Launch Vehicle-1 (KSLV-1) blasted off from the Naro Space Center at 16:00, marking a major milestone for South Korea's space program. This launch was the country's first successful attempt at putting a satellite into orbit using its own rocket. LLM likelihood: 1.00 	At about 1600 a.m. the Korea Space Launch Vehicle-1 (KSLV-1) blasted off from the Naro Space Center, marking the first successful attempt by Korea to launch a satellite into space. The KSLV-1, also known as the "Naro," was a rocket jointly developed by the Korea Aerospace... LLM likelihood: 0.89 	KSLV-1 just launched from Naro Space Center at 4 PM, big win for South Korea's space program. First time they successfully put a satellite in orbit with their own rocket. Developed with KARI and Roscosmos, they failed tries in 2009 and 2010. Now they're part of the elite club that... LLM likelihood: 0.04 
Disguised as police, they broke through a fence on Monday evening and broke into the cargo of a Swiss-bound plane to take the valuable items. The audacious heist occurred at an airport in a small European country, leaving authorities baffled and airline officials in shock... LLM likelihood: 1.00 	An audacious robbery was perpetrated in a small European country on the eve of the International holiday of St. Jean. The robbers, disguised as police, made their operations so carefully planned as to make the whole operation look like an official investigation. In this way they ... LLM likelihood: 0.98 	Criminals dressed as cops busted through a fence and hit the cargo of a plane headed to Switzerland on Monday night. The heist went down at a small European airport, totally throwing off the authorities and airline people. They made it look like a legit police operation... LLM likelihood: 0.15 

Figure 5: Paraphrased sentences by Dipper and our CoPA. We use Fast-DetectGPT to provide the LLM likelihood.

paraphrase corpus (Wieting and Gimpel, 2018), to measure the semantic discrepancy. We align with Dipper and consider the semantics being preserved if the P-SP score exceeds the average real-human paraphrase score of 0.76. Moreover, we provide *more analysis including text perplexity, GPT-4 assisted and human evaluation in Section H*.

Implementation Details. For attack hyperparameters, we set the contrast intensity $\lambda = 0.5$ and the clipping factor $\alpha = 1e^{-5}$. Unless stated otherwise, we employ a single paraphrasing iteration. We employ Qwen2.5-72B-Instruct (Qwen, 2024) as the paraphraser. Due to page limits, we provide results paraphrased by more LLMs in Section C. More details about the human and machine-like prompts are in Appendix B.

4.2 Performance Evaluation

We test machine texts generated by GPT-3.5-turbo and present results on three datasets in Table 1.

Attack Effectiveness. By conducting a self-introspective correction on token distributions, CoPA remarkably enhances the attack over baseline attacks, *e.g.*, an average improvement of 30.55% in fooling text detectors across three datasets. Although Dipper demonstrates satisfactory perfor-

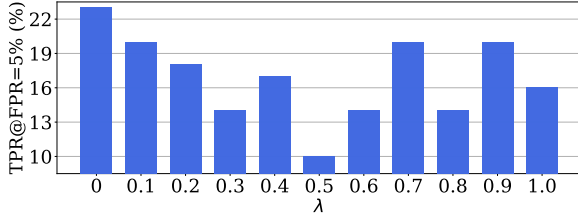
mance against several detectors, it becomes significantly less effective when facing more advanced algorithms such as FastDetectGPT. In contrast, our method consistently exhibits impressive attack efficacy across various text detectors. Notably, while Raidar-A and our method employ the same LLM as the paraphraser, CoPA greatly outperforms Raidar-A, validating the effectiveness of our designed prompt and contrastive paraphrasing mechanism.

As for the text quality of rewritten sentences, we demonstrate that CoPA achieves an average semantic similarity score exceeding 90% across various datasets, confirming that our method effectively preserves semantic fidelity during rewriting. While Raidar-A exhibits greater text similarity, its attack effectiveness remains considerably limited. As a comparison, CoPA achieves both excellent attack effectiveness and semantic consistency.

Visualization of Rewritten Texts. We provide examples of rewritten texts before and after the paraphrasing in Figure 5. As expected, the rewritten sentences maintain semantic consistency while exhibiting richer and diverse human-like expressions. This underpins our success in fooling text detectors and presents a practical attack method.

Table 2: Attack Performance (at 5% FPR) of texts generated by more source LLMs based on XSum dataset.

Model	Attack	Sim	Defense								Avg.
			LogRank	DetectGPT	DNA-GPT	Fast-DetectGPT	Raidar	TOCSIN	RoBERTa	R-Detect	
GPT-4	No Attack	-	30.00	6.00	35.33	51.67	24.17	73.33	32.67	46.00	37.40
	Dipper	91.33	8.67	0.67	30.67	64.33	20.83	64.67	78.00	37.67	38.19
	Raidar	100.00	35.00	9.50	34.67	68.33	20.83	82.00	47.33	60.50	44.77
	Ours	94.67	2.00	0.67	18.67	15.33	10.83	20.00	20.67	6.83	11.88
Claude 3.5	No Attack	-	42.67	21.67	24.67	50.00	36.67	70.00	19.33	30.67	36.96
	Dipper	82.00	18.17	0.33	20.67	38.67	0.00	36.67	77.33	40.33	29.02
	Raidar	100.00	46.83	18.77	15.33	41.33	41.64	66.00	20.67	38.00	36.07
	Ours	98.00	1.33	0.67	6.67	4.00	0.00	8.00	7.33	1.33	3.67

Figure 6: Comparison of detection accuracy on the first 50 samples from XSum under different values of λ against Fast-DetectGPT (Bao et al., 2024).

4.3 Attacks on More Source LLMs.

In addition to GPT-3.5-turbo, we also consider the machine texts generated by recently prevalent LLMs, including GPT-4 (Achiam et al., 2023) and Claude-3.5 (Anthropic, 2024). Besides, we provide results on GPT-4o (Achiam et al., 2023) and Gemini-1.5 Pro (Team et al., 2024) in Appendix C.

Table 2 demonstrates that our method continues to achieve excellent attack efficacy and semantic similarity across various LLM-generated texts, greatly outperforming the SOTA method Dipper. For detection of GPT-4 texts under Fast-DetectGPT, Dipper even increases the likelihood to be classified as machine-generated than the *No Attack* baseline. In contrast, CoPA stably achieves outstanding fooling rates across various source models, confirming the robustness of the proposed contrastive paraphrase. Another observation is that the detection performance on clean texts generated by more recent models is significantly reduced. The decline may stem from the ability of advanced LLMs to generate varying sentences that are more challenging to detect, underscoring the urgent need for more reliable detection systems.

4.4 Ablation Study

We then investigate the effect of different factors. More ablation studies are in Appendix C.

Impact of contrastive coefficient λ . During decoding, the hyperparameter λ serves as a criti-

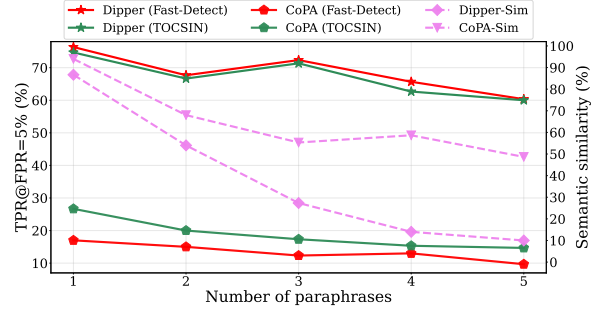


Figure 7: Performance under different numbers of paraphrases against Fast-DetectGPT (Bao et al., 2024) and TOCSIN (Ma and Wang, 2024) on samples generated by GPT-3.5-turbo from the XSum dataset. The dashed lines describe semantic similarity.

cal regulation factor for the contrast strength. As shown in Figure 6, positive values of λ consistently enhance the fooling rates relative to $\lambda = 0$, again confirming the effectiveness of our contrastive paraphrasing mechanism. Note that CoPA attains optimal performance at $\lambda = 0.5$, which is then chosen as the default setting for our experiments. Notably, the general trend of TPR over the λ roughly aligns with our preceding theoretical analysis, verifying the rationality of our established theory.

Impact of Multiple Paraphrases. Each LLM-generated text is rewritten only once in our former experiments. We further analyze the influence of multiple rewrites on the results. As shown in Figure 7, increasing the number of rewrites generally strengthens attack effectiveness. However, the performance gain is limited against two advanced defenses, and the semantic similarity of Dipper-rewritten texts sharply drops as the iterations increase. These factors reduce the utility of using multiple paraphrases to improve the attack (Sadasivan et al., 2023). Also, this again highlights the superiority of our method, which achieves outstanding fooling rates via only a single paraphrasing.

5 Conclusion

This paper proposes CoPA, a simple yet highly effective paraphrasing attack against AI-generated text detectors. CoPA constructs a machine-style token distribution as a negative contrast for reducing linguistic biases of LLMs and facilitating the generation of richer and more diverse sentences. Through both theoretical analysis and experimental validation, we fully demonstrate the superiority of the proposed method across various scenarios. We envision CoPA as a powerful tool for auditing the robustness of detection systems, inspiring future development of more robust detection algorithms.

Limitations

While our method avoids the overhead of training a dedicated paraphraser by leveraging an off-the-shelf LLM, the contrastive paraphrasing mechanism requires two forward passes to construct the contrastive token distribution, bringing additional latency during next-token prediction. This may limit the practicality of the proposed attack in real-time applications. Moreover, using off-the-shelf LLMs to paraphrase texts can lead to the semantics of output texts deviating from the original sentence, which may require multiple generations to maintain the semantic similarity. Besides, although human evaluation results in Appendix H indicate that CoPA-paraphrased texts are preferred over those from Dipper, we did not systematically account for the detailed linguistic backgrounds of evaluators and may introduce bias. A more comprehensive human study is needed to validate the general quality of the output sentences. Finally, this work follows prior studies and focuses exclusively on English text. Extending the contrastive paraphrasing framework to other languages, such as Chinese and Spanish, would be valuable for its broader applicability.

Ethical Statement

This paper presents a novel method aimed to advance the research field of LLM-generated text detection. Note that all experiments are conducted within controlled laboratory environments. We do not expect the proposed method to serve as a powerful tool for potential adversaries but to raise society’s broader awareness of the vulnerability of current AI-text detectors. Also, the exceptional attack performance highlights the practical limitations of current detectors. Researchers of the open-source community are encouraged to conduct

stress tests on their detectors against the proposed attack, based on which future studies can develop more robust stronger detectors. **Furthermore, we conduct a preliminary study to alleviate the proposed threat via an adaptive defense that adversarially trains a Roberta-based detector using texts paraphrased by CoPA in Section F.**

All the codes, models, and datasets used in this study are consistent with their intended use and comply with the MIT License. To promote further research, we will open-source our paraphrasing tool along with the related code, model, and data.

Acknowledgement

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